

LECTURE-26

MATHEMATICS 1 (15B11MA111)

DIFFERENTIAL EQUATION WITH CONSTANT COEFFICIENT-PARTICULAR INTEGRAL

Differential Equation CO[C105.5]

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- R.K. Jain and S.R.K. Iyenger, "Advanced Engineering Mathematics" fifth edition, Narosa publishing house, 2016.

Particular Integral

Consider the linear non homogeneous differential equation with constant coefficients

$$L(y) = a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = r(x)$$

or

$$L(y) = (a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n) y = F(D) y = r(x),$$

where $F(D) = a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n$ and $a_0, a_1, \dots, a_n$ are constants. Then the particular integral is given by

$$PI = y_p(x) = [F(D)]^{-1} r(x) = \frac{1}{F(D)} r(x).$$

Rules for Finding the Particular Integral

Case 1. When $r(x) = e^{\alpha x}$

If $r(x) = e^{\alpha x}$, then the particular Integral is given as

$$y_p(x) = \frac{1}{F(\alpha)} e^{\alpha x}, \quad F(\alpha) \neq 0.$$

Since

$$D e^{\alpha x} = \alpha e^{\alpha x}, \quad D^2 e^{\alpha x} = \alpha^2 e^{\alpha x} \dots, \quad D^n e^{\alpha x} = \alpha^n e^{\alpha x}$$

Therefore

$$\begin{aligned} (a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n) e^{\alpha x} \\ = (a_0 \alpha^n + a_1 \alpha^{n-1} + \dots + a_{n-1} \alpha + a_n) e^{\alpha x} \end{aligned}$$

Or

$$\Rightarrow F(D) e^{\alpha x} = F(\alpha) e^{\alpha x}.$$

Cont...

Operating $[F(D)]^{-1}$ on both sides of the above equation and dividing by $F(\alpha)$, we get

$$PI = y_p(x) = \frac{1}{F(D)} e^{\alpha x} = \frac{1}{F(\alpha)} e^{\alpha x}, \text{ provided } F(\alpha) \neq 0.$$

Example: Find the particular integral of the following differential equation

$$y'' - 2y' - 3y = 3e^{2x}.$$

Solution: Given differential equation $(D^2 - 2D - 3)y = 3e^{2x}$ is a non homogeneous linear differential equation. The particular integral is given as

$$y_p(x) = \frac{1}{F(D)} 3e^{2x} = \frac{1}{D^2 - 2D - 3} 3e^{2x} = \frac{1}{2^2 - 2 \cdot 2 - 3} 3e^{2x} = -e^{2x}.$$

If $F(\alpha) = 0$, the above rule fails and we have

$$y_p(x) = \frac{x}{F'(\alpha)} e^{\alpha x},$$

provided $F'(\alpha) \neq 0$.

If $F(\alpha) = 0 = F'(\alpha)$. Then the particular integral

$$y_p(x) = \frac{x^2}{F''(\alpha)} e^{\alpha x},$$

provided $F''(\alpha) \neq 0$.

Similarly If $F(\alpha) = 0 = F'(\alpha) = F''(\alpha) = \dots = F^{n-1}(\alpha)$. Then the particular integral

$$y_p(x) = \frac{x^n}{F^n(\alpha)} e^{\alpha x},$$

provided $F^{(n)}(\alpha) \neq 0$.

Example: Find the general Solution of the following differential equation

$$(D^2 - 2D + 1)y = e^x.$$

Solution: The auxiliary equation is given as $m^2 - 2m + 1 = 0$, on solving we get $m = 1, 1$. Hence the complimentary function is given by

$$CF = (c_1 + c_2x)e^x.$$

Now the particular integral is given as

$$PI = \frac{1}{D^2 - 2D + 1}e^x = x \frac{1}{2D - 2}e^x = x^2 \frac{1}{2}e^x = \frac{x^2}{2}e^x,$$

since $F(1) = F'(1) = 0$. The general solution is therefore

$$y(x) = CF + PI = (c_1 + c_2x)e^x + \frac{x^2}{2}e^x.$$

CASE 2: When $r(x) = \cos(\alpha x + \beta)/\sin(\alpha x + \beta)$

The particular integral of $F(D)y = \cos(\alpha x + \beta)/\sin(\alpha x + \beta)$, is given as

$$y_p(x) = \frac{1}{F(D^2)} \cos(\alpha x + \beta) \text{ or } \sin(\alpha x + \beta) = \frac{1}{F(-\alpha^2)} \cos(\alpha x + \beta) / \sin(\alpha x + \beta)$$

replacing D^2 by $-\alpha^2$ in $F(D^2)$ provided $F(-\alpha^2) \neq 0$.

If replacing D^2 by $-\alpha^2$ we get $F(-\alpha^2) = 0$, then the above rule fails and we have the particular integral

$$y_p(x) = \frac{x}{F'(-\alpha^2)} \cos(\alpha x + \beta) / \sin(\alpha x + \beta),$$

provided $F'(-\alpha^2) \neq 0$.

If $F(-\alpha^2) = 0 = F'(-\alpha^2)$. Then the particular integral

$$y_p(x) = \frac{x^2}{F''(-\alpha^2)} \cos(\alpha x + \beta) / \sin(\alpha x + \beta),$$

provided $F''(-\alpha^2) \neq 0$.

Similarly if $F(-\alpha^2) = 0 = F'(-\alpha^2) = F''(-\alpha^2) = \dots = F^{n-1}(-\alpha^2)$.

Then the particular integral

$$y_p(x) = \frac{x^n}{F^{(n)}(-\alpha^2)} \cos(\alpha x + \beta) / \sin(\alpha x + \beta),$$

provided $F^{(n)}(-\alpha^2) \neq 0$.

Example: Find the particular integral of the following differential equation

$$(D^2 + 4)y = \sin x + \cos 2x.$$

Solution: Here $F(D^2) = D^2 + 4$, therefore the particular integral is given as

$$\begin{aligned} PI &= \frac{1}{D^2 + 4}(\sin x + \cos 2x) \\ &= \frac{1}{D^2 + 4}\sin x + \frac{1}{D^2 + 4}\cos 2x \\ &= \frac{1}{-1 + 4}\sin x + x \frac{1}{2D}\cos 2x \quad \text{since } F(-2^2) = 0 \\ &= \frac{1}{3}\sin x + \frac{x}{2} \frac{\sin 2x}{2} \\ &= \frac{1}{3}\sin x + \frac{x}{4}\sin 2x. \end{aligned}$$

Practice Questions

1: Find the general solution of the following differential equation

$$y'' + y' + y = (1 - e^x)^2.$$

Ans: $y(x) = A \cos\left(\frac{\sqrt{3}}{2}x\right) + B \sin\left(\frac{\sqrt{3}}{2}x\right) + 1 - \frac{2}{3}e^x + \frac{1}{7}e^{2x}.$

2: Find the particular integral of the following differential equation

$$(D - 2)^2 y = 8(e^{2x} + \sin 2x).$$

Ans: $y_p = 4x^2 e^{2x} + \cos 2x.$

3: Find the particular integral of $(D^3 + 1)y = \cos(2x - 1).$

Ans: $y_p = \frac{1}{65} [\cos(2x - 1) - 8\sin(2x - 1)].$

Thank
you !